U of M Carlson Analytics Lab

Bridging the Gap: Validating Route Execution Through GPS Analytics

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4MATIV

AGENDA

01

Problem Statement

Our Journey

Turn imperfect data into reliable insights

01
IDEAL WORLD

In an ideal world,
GPS data accurately reflects route execution

02
REALITY

But in reality,
GPS data is often unreliable, making it hard to verify whether routes were actually executed as planned

03
OUR GOAL

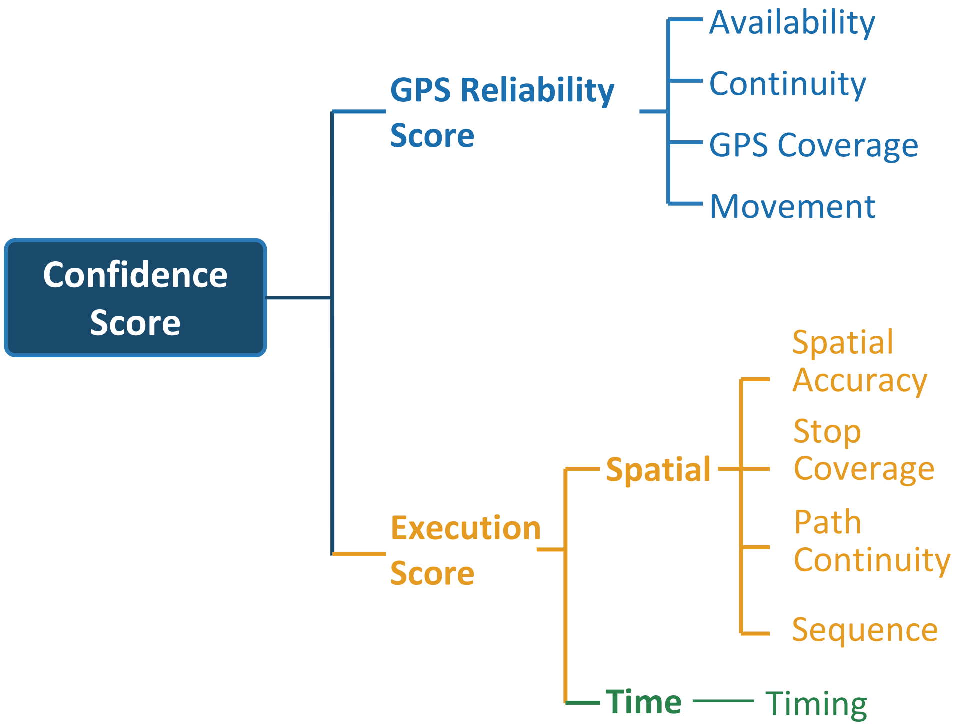
Our goal
is to reliably assess whether routes were executed as planned, even with imperfect GPS data

02

Solution Map

Confidence Score = GPS Quality × How Well the Route Was Executed

Bad GPS ≠ Bad Execution — the score separates data noise from real operational issues



FORMULA

Confidence Score =

GPS Reliability Score × (

Spatial Accuracy + Stop Coverage + Path Continuity + Sequence + Timing)

PERFECT SCORE EXAMPLE

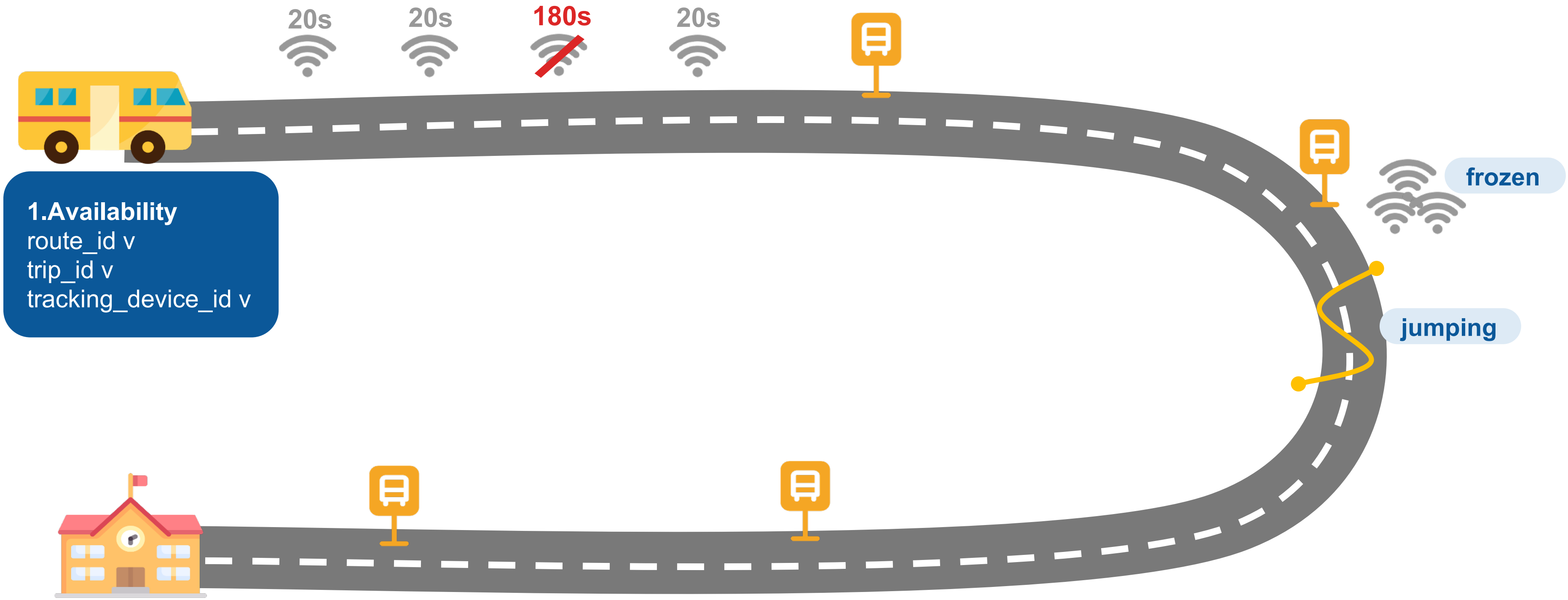
10 =

1 × (**2** + **2** + **2** + **2** + **2**)

GPS = 1 pts (perfect quality) |
each component = 2 pts | max total = 10

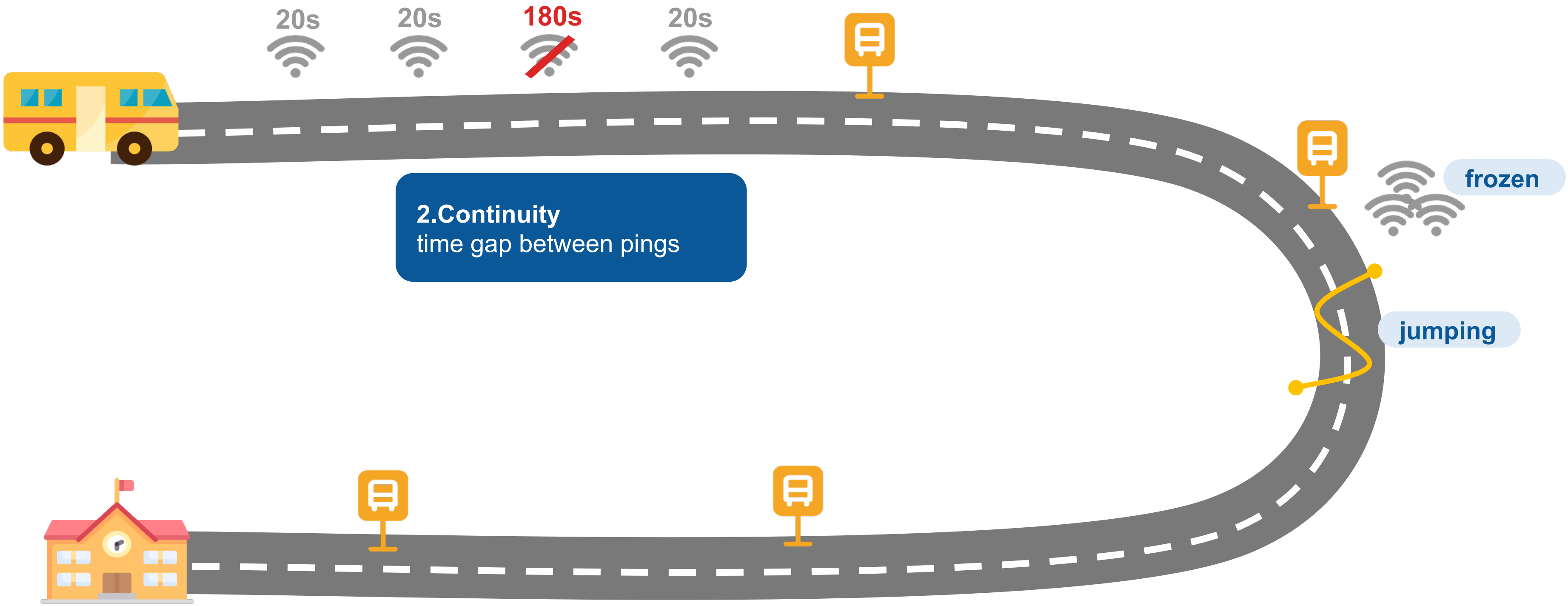
● GPS Reliability ● Spatial ● Timing

GPS Reliability Score

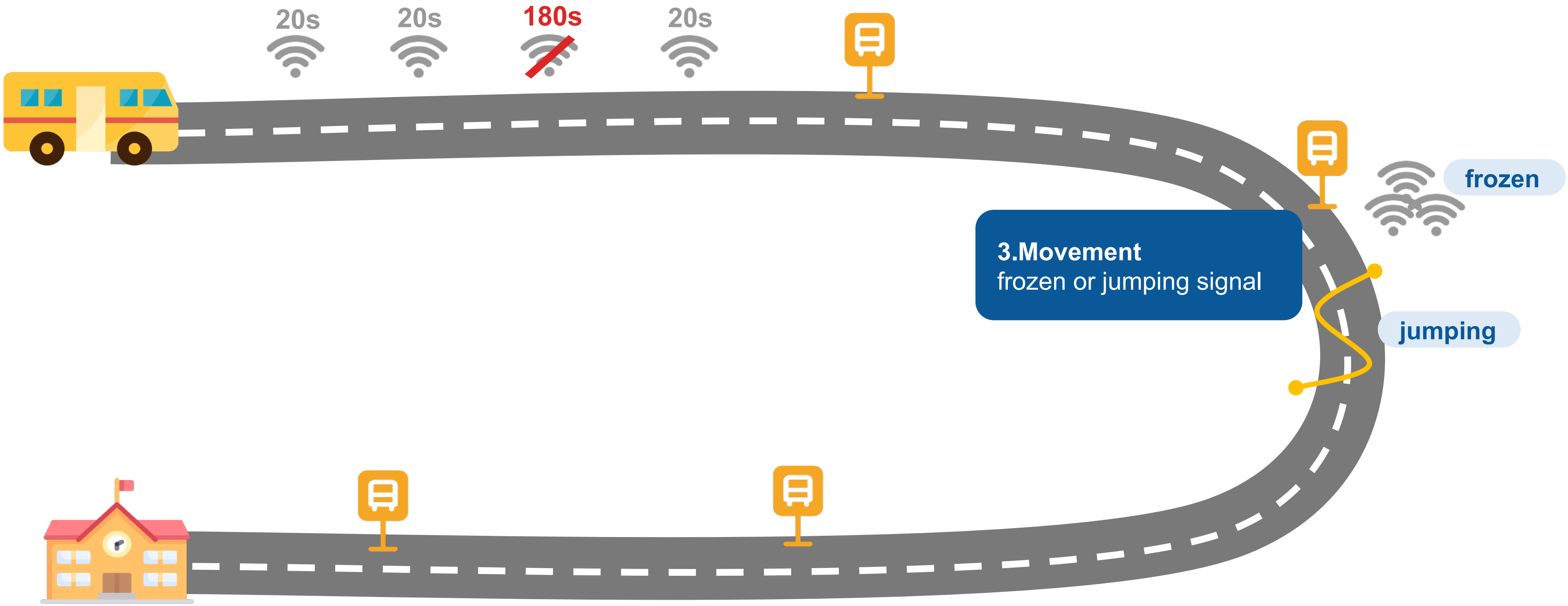


1.Availability
route_id v
trip_id v
tracking_device_id v

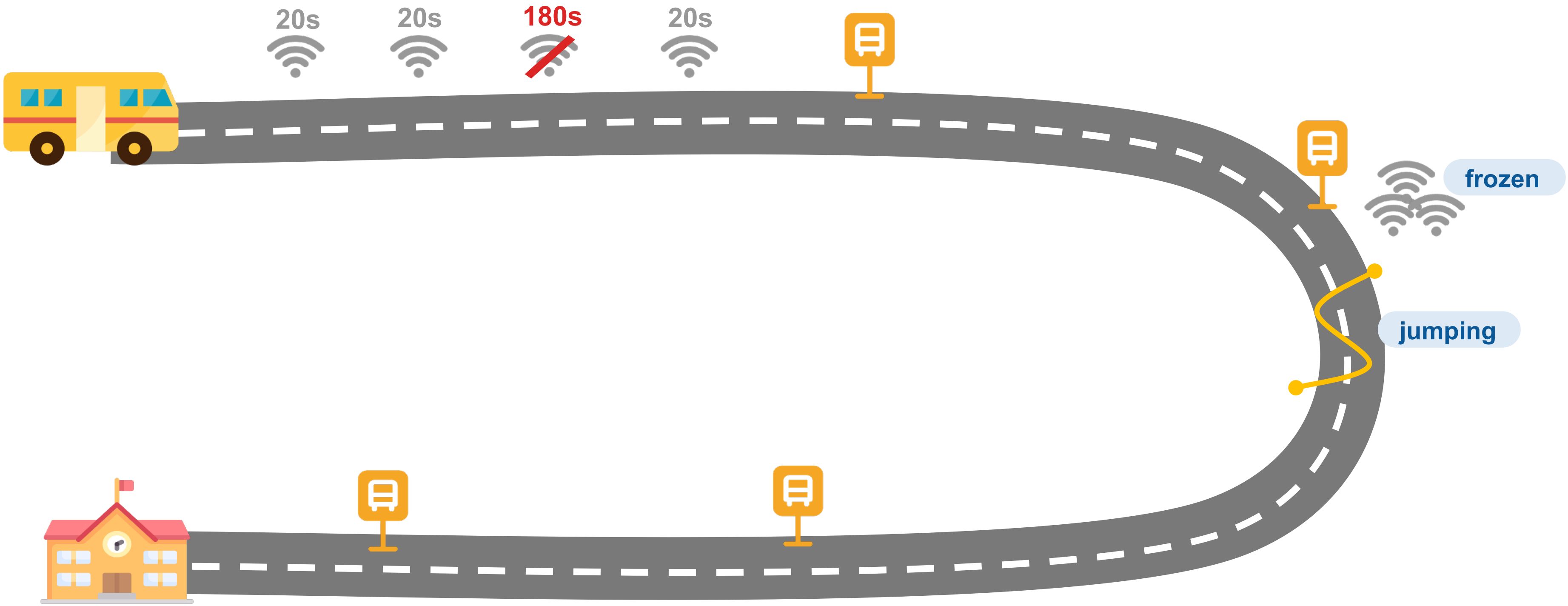
GPS Reliability Score



GPS Reliability Score

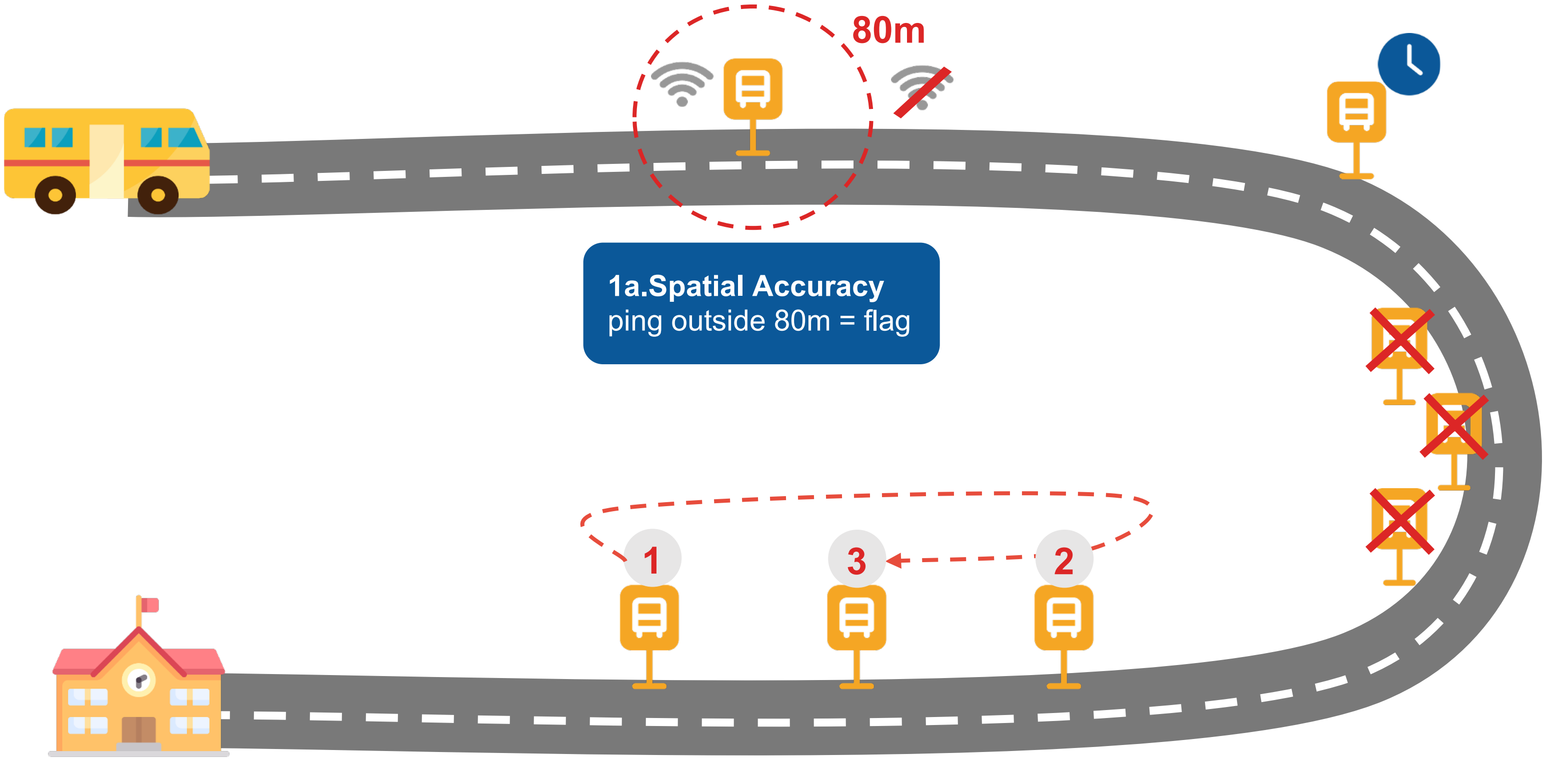


GPS Reliability Score

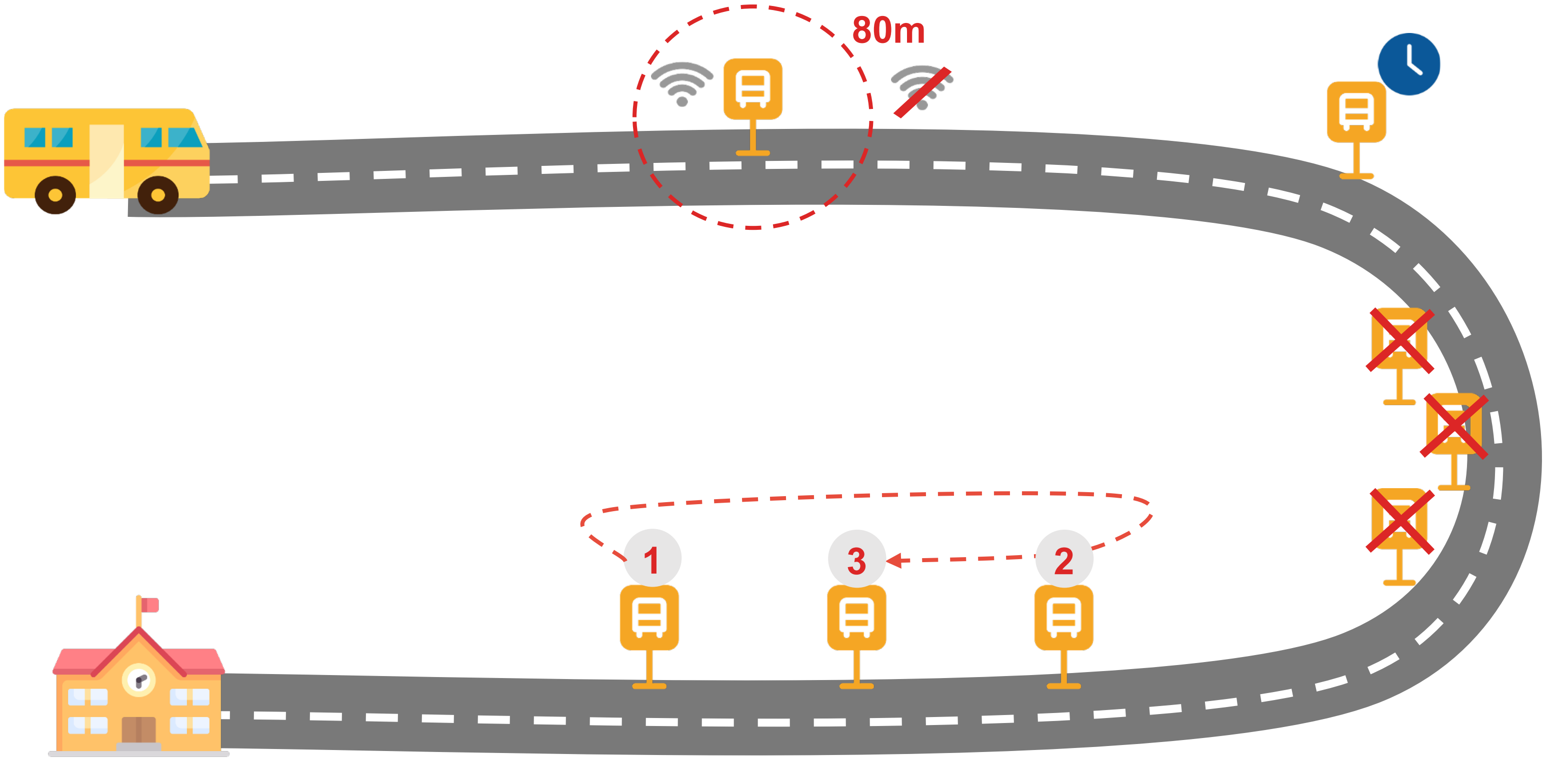


4.GPS Coverage
actual pcts vs. expected pcts
30 pcts act 30 mins / 20s = 90 pcts

Execution Score (Spatial + Timing)

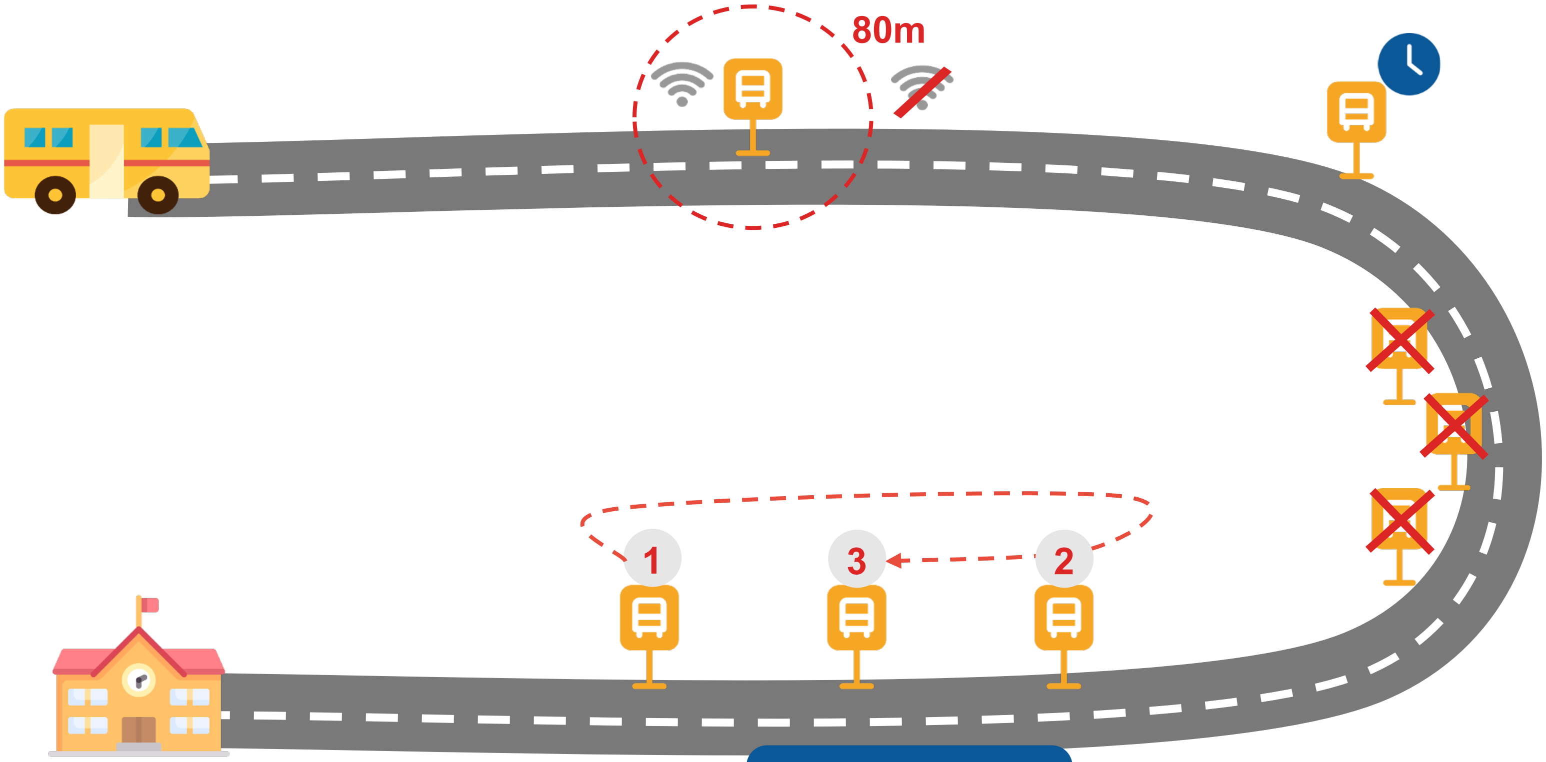


Execution Score (Spatial + Timing)



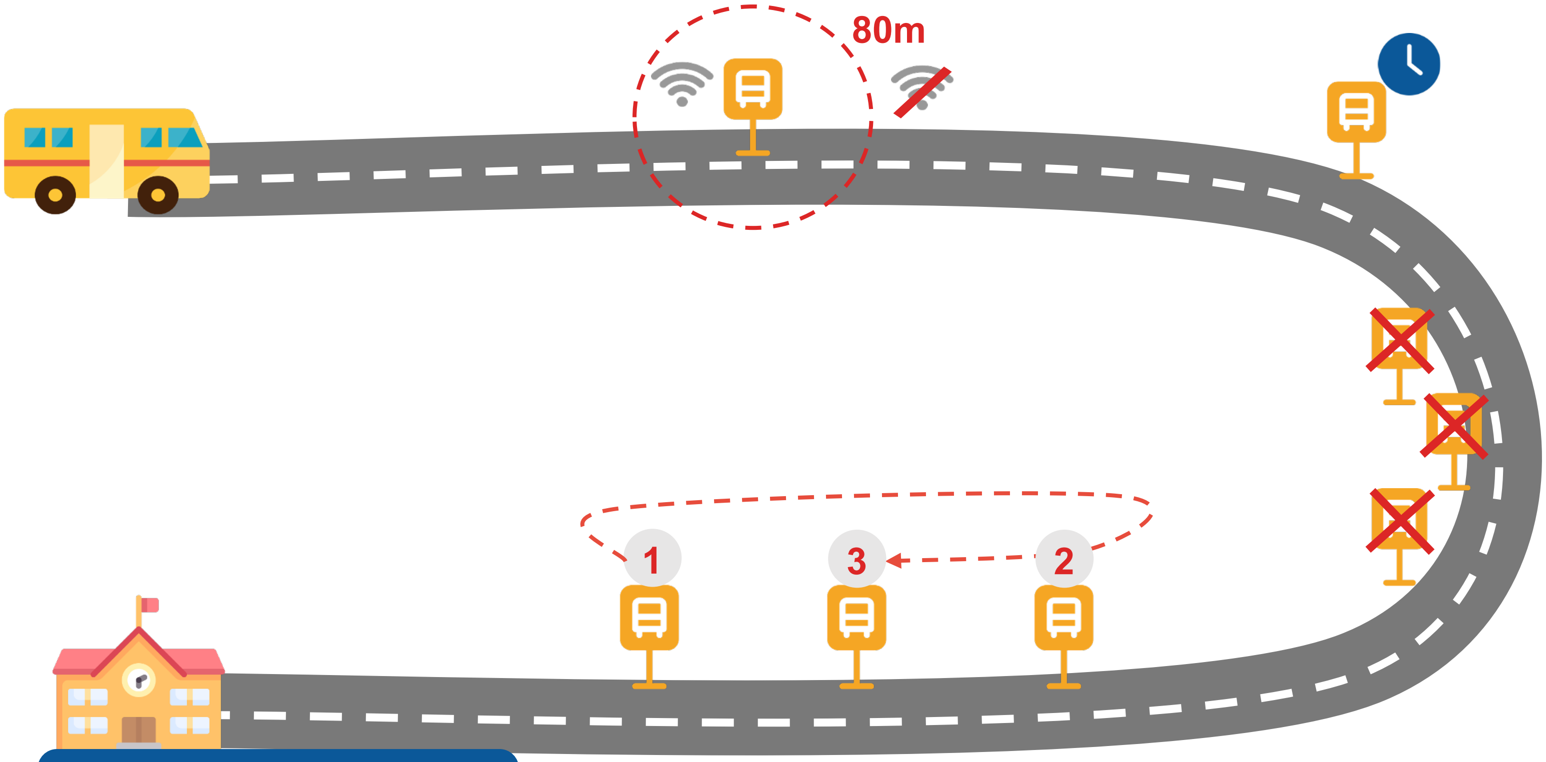
1b.Continuity
missing ping chain

Execution Score (Spatial + Timing)



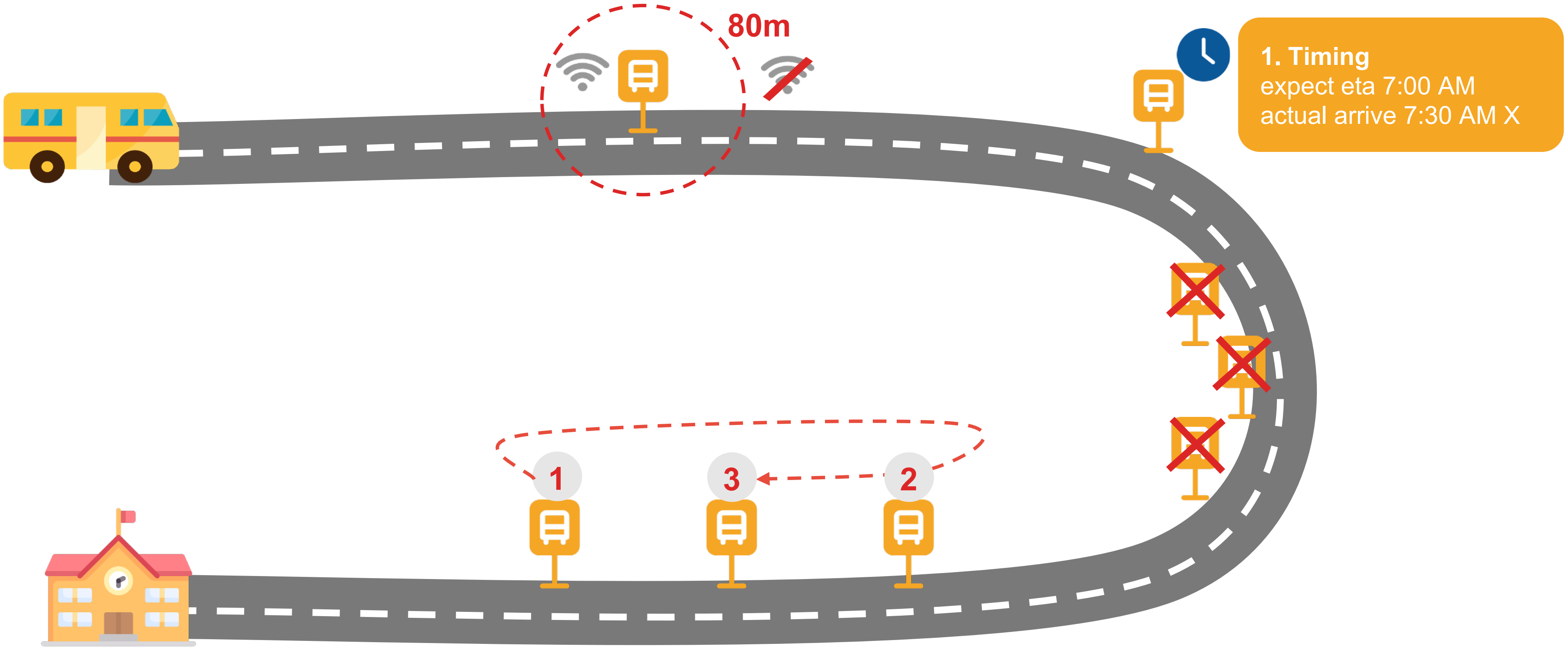
1c.Sequence
wrong visiting order

Execution Score (Spatial + Timing)



1d.Stop Coverage
act matched stop / planned stop

Execution Score (Spatial + Timing)



Indicators of GPS Reliability

Indicator	What it checks	How it's computed	Rule (data-driven)	Outcome
Availability (Hard Check)	Is GPS usable?	Device + GPS data + system trigger	Fail if ANY: • no device (missing tracking_device_id) • no GPS data (0 points) • system failure (end_trigger = no_gps_data ~3% trips)	Fail → Unreliable
Continuity (Signal Stability)	Is signal consistent?	gap_sec → provider thresholds → trip ratios	Point-level (provider-based): • good ≤ p90 (normal interval ~30–60s) • weak > p90 (top ~10% gaps) • bad > p99 (extreme gaps) Trip-level (dataset-based): • bad → gap_bad_ratio > 0.036 (~top 5% trips) • weak → gap_weak_ratio > 0.025 (~top 10% trips) • good → rest	Good / Weak → Pass Bad → Degraded
GPS Coverage	Is trip fully captured?	actual / expected points (duration ÷ provider freq)	• good ≥ 0.8 (near-complete) • weak: 0.6–0.8 (partial) • bad < 0.6 (~bottom 20% trips) (capped at 1.5 to remove noise)	Good / Weak → Pass Bad → Degraded
Movement	Is movement realistic?	static_ratio + jump_ratio	Static (distribution-based): • weak > P75 (top 25% static) • bad > P90 (top 10% frozen) Jump (override): • weak > 0.09 (~P90) • bad > 0.16 (~P95)	Good / Weak → Pass Bad → Degraded

Indicators of Execution Score

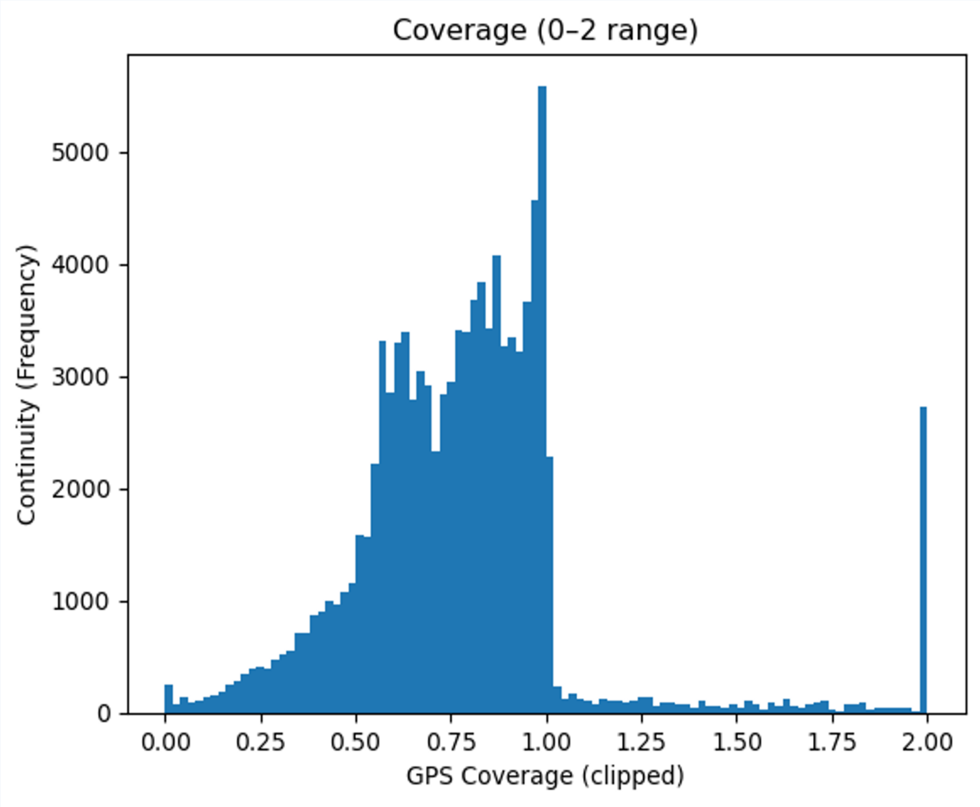
Indicator	Purpose	Metric / Formula	Anomaly Flag / Rule	Score
Stop Coverage	Measure stop completion	# actual matched stops / planned stops	zero coverage = 0, low coverage < 0.5	rate = 1 → 2 pts 0.5 < rate < 1 → 1 pts rate < 0.5 → 0 pts
Spatial Accuracy	Measure stop proximity	% matched stops inside 80m	low_spatial_accuracy : <50%	rate = 1.0 → 2 pts 0.8 ≤ rate < 1.0 → 1.5 pts 0.5 ≤ rate < 0.8 → 1 pts < 0.5 → 0 pts
Path Continuity	Measure dropout severity and gap fragmentation	Longest Missing Run & Jump Events Count	Good: one metric < 25th pct and the other < 50th pct Poor: one metric > 75th pct and the other > 50th pct Fair: otherwise	Good → 2 pts Partial → 1 pts Poor → 0 pts
Sequence	Measure sequence adherence	% actual vs planned sequence	Good (2): ≥ 75% stops in order Partial (1): 33 – 74% stops in order Poor (0): < 33% stops in order	Good → 2 pts Partial → 1 pts Poor → 0 pts

Indicator	Purpose	Metric / Formula	Anomaly Flag / Rule	Score
Timing	Check if the executions on time as planned	# Completed_diff # Departed_diff	Early : diff < -10 On time : -10 ≤ diff ≤ +10 Late : +10 < cdiff ≤ +20 Very late : diff > +20	On time & Early → 2 pts Late → 1 pts Very Late → 0 pts

03

GPS Reliability Score

GPS Data Appears Complete - But Isn't



What we observe:

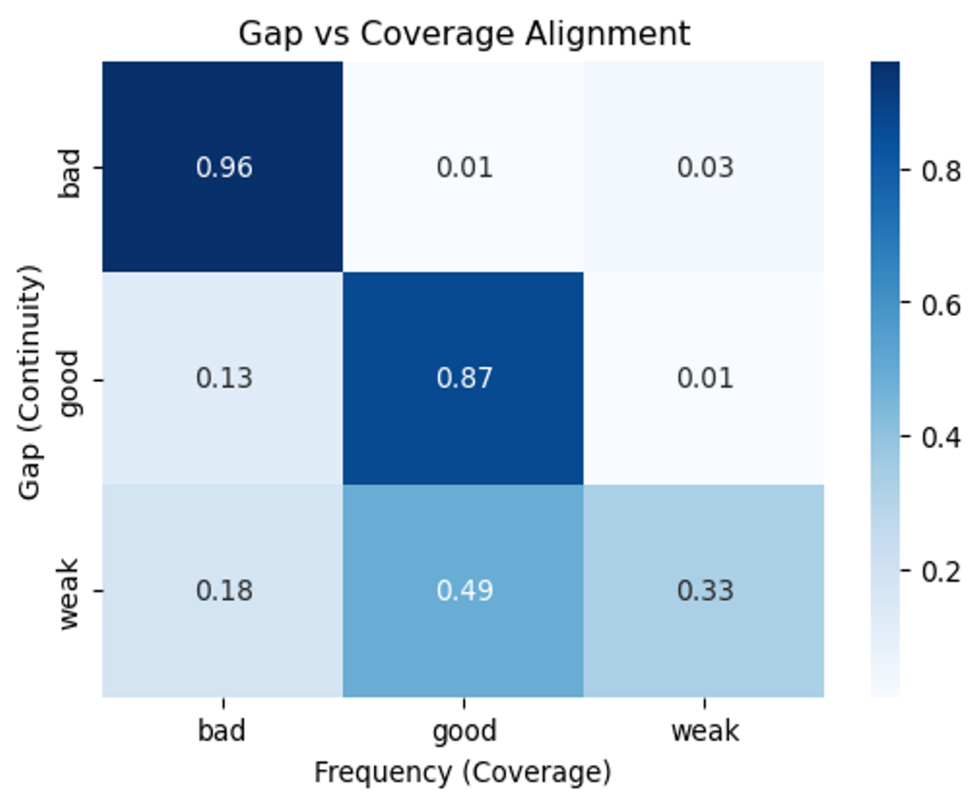
- Most trips show **moderate to high GPS coverage (median ~0.79)**
- However, there is a **significant tail of low coverage (<0.5)**

Implication:

- GPS data is **available**, but not consistently **complete**
- Many trips have **partial visibility of actual movement**

GPS presence does not guarantee full trip visibility

Stable GPS Signals Do Not Guarantee Reliable Data



Key Observations

When GPS is bad, it's clearly bad
→ ~96% of poor continuity also have poor coverage

But "good" GPS isn't always reliable
→ ~14% of trips with good continuity still have incomplete coverage

Overall, coverage is inconsistent
→ Only ~87% of trips with good continuity have complete GPS data

Insight:

Even when GPS looks stable, a portion of trips still have incomplete visibility, making raw GPS unreliable for validation.

GPS Reliability

Measures the reliability of GPS data based on availability and signal quality

WHY IT MATTERS

Ensures GPS data is reliable enough to confirm whether the route was actually followed

HOW WE MEASURE IT

Availability (Hard Check)
Device present + GPS data available + no system failure

Quality Signals
Continuity · Coverage · Movement (any "bad" flag → GPS is degraded)

CATEGORY & SCORE

Category	Description	Share	Score
Good	Reliable **Availability OK AND no bad quality flags**	68.7%	1 pt
Partial	Unknown **Availability OK AND any bad flag (gap, freq, or frozen)**	19.0%	0.5 pt
Poor	Unreliable **Availability fails (invalid ID, missing data, or invalid trigger)**	11.7%	0 pt

GPS Reliability Demo

Metric-level comparison and final reliability scoring

TRIP RECORD 1083730	
RELIABLE	
RELIABILITY SCORE	1
Device	VALID
Data	PRESENT
System	VALID
Continuity	WEAK
GPS Coverage	WEAK
Movement	GOOD

TRIP RECORD 1216865	
UNKNOWN	
RELIABILITY SCORE	0.5
Device	VALID
Data	PRESENT
System	VALID
Continuity	WEAK
GPS Coverage	BAD
Movement	GOOD

TRIP RECORD 1060337	
UNRELIABLE	
RELIABILITY SCORE	0
Device	VALID
Data	MISSING
System	VALID
Continuity	GOOD
GPS Coverage	GOOD
Movement	UNKNOWN

04

Execution Score

Confidence Score = GPS Quality × How Well the Route Was Executed

- **Confidence Score**
  - **GPS Reliability Score**
    - Availability
    - Continuity
    - GPS Coverage
    - Movement
  - **Execution Score**
    - **Spatial**
      - Spatial Accuracy
      - Stop Coverage
      - Path Continuity
      - Sequence
    - **Time**
      - Timing

● GPS Reliability ● Spatial ● Timing

Path Continuity

Measures whether missing GPS data creates only minor gaps or breaks the route trace enough to weaken execution validation.

WHY IT MATTERS

Path continuity shows whether the route trace remains continuous enough to validate that **the assigned route was actually followed**

HOW WE MEASURE IT

Longest Missing Run

Longest block of consecutive missed planned stops

Jump Events Count

Number of separate missing-gap segments

CATEGORY & SCORE

Category	Criteria	Share
Good	**(run ≤ 1 and jump ≤ 3)** **or (run ≤ 2 and jump ≤ 1)** Low dropout pattern	28.4% 2 pts
Partial	**All other cases** Mixed or moderate gaps	36.9% 1 pts
Poor	**(run ≥ 4 and jump ≥ 3)** **or (run ≥ 2 and jump ≥ 4)** Fragmented dropout	34.7% 0 pts

Sequence Adherence

Are stops being visited in the right order?

WHY IT MATTERS

Routes/trips sequence adherence reflects **operational discipline** and **planning compliance**.

HOW WE MEASURE IT

Actual Sequence vs. Planned Sequence

Compare each trip's actual stop sequence against planned for that route and direction. Use *Longest Common Subsequence* to capture which stops were visited and whether they were visited in the correct relative order

CATEGORY & SCORE

Category	Description	Share	Score
Good	**Strong Adherence** ≥75% of planned stops done in order	72.5%	2 pts
Partial	**Noticeable Gaps but not a Complete Mismatch** 33–74% in order	27.5%	1 pts
Poor	**Essentially Off-route** <33% in order	0.05%	0 pts

**Removed trips with zero and only one stops*

Timing – Stop Level

Measures how each stop deviates from planned arrival and departure times

WHY IT MATTERS

Stop-level timing is the most granular view of schedule adherence. Comparing arrival vs. departure reveals **where time is lost at stops** — which feeds directly into trip-level scoring and vendor accountability.

HOW WE MEASURE IT

Stop Arrival Time Difference

Actual – Planned arrival time at each stop (completed_diff)

Stop Departure Time Difference

Actual – Planned departure time at each stop (departed_diff)

Category	Threshold	ARRIVAL (completed_diff)		DEPARTURE (departed_diff)	
On Time	≤ +10 min	83.9%	70,811 stops	79.7%	61,746 stops
Late	+10 to +20 min	7.4%	6,238 stops	12.0%	9,261 stops
Very Late	> +20 min	2.2%	1,814 stops	4.8%	3,688 stops
Early	< −10 min	6.5%	5,509 stops	3.5%	2,731 stops

Timing – Trip Level Scoring

Aggregates stop-level deviations to score each trip's overall schedule adherence

WHY IT MATTERS

Trips provide a more **holistic view** of execution performance by capturing the full journey rather than isolated stop-level events. Aggregating to the trip level allows us to evaluate overall execution quality, making it a more appropriate unit for calculating the Confidence Score.

HOW WE MEASURE IT

Trip Average Time difference

Mean of all stop-level completed_diff values within the trip

Timing Score

0-2 pts assigned per trip based on avg_diff category

CATEGORY & SCORE

On Time (& Early)

avg_diff ≤ +10 min (incl. Early trips)

Trip executed within acceptable window

93.0% / 2 pts

7,792 trips

Late

avg_diff +10 to +20 min

Trip runs behind — partial service delivery issue

5.3% / 1 pt

445 trips

Very Late

avg_diff > +20 min

Severe delay — direct flag for execution review

1.7% / 0 pts

141 trips

05

Score Demo

Execution Score

Trip ID 1115526

Stop Coverage	Spatial Accuracy	Path Continuity	Sequence	Timing
Perfect	Good	Partial	Good	Very Late
SCORE	SCORE	SCORE	SCORE	SCORE
2	1.5	1	2	0
/ 2 pts	/ 2 pts	/ 2 pts	/ 2 pts	/ 2 pts

TOTAL EXECUTION SCORE

= **6.5 / 10 pts**

Confidence Score

Trip ID 1115526

GPS Reliability Score		Execution Score		Confidence Score
1	×	6.5	=	6.5
✓ Reliability		Moderate performance		Review
/ 1 pts		/ 10 pts		/ 10 pts

Confidence Score Interpretation (Trip Level)

Flag < 5

Data quality issues — do not use for driver assessment

REVIEW 5 – 7.5

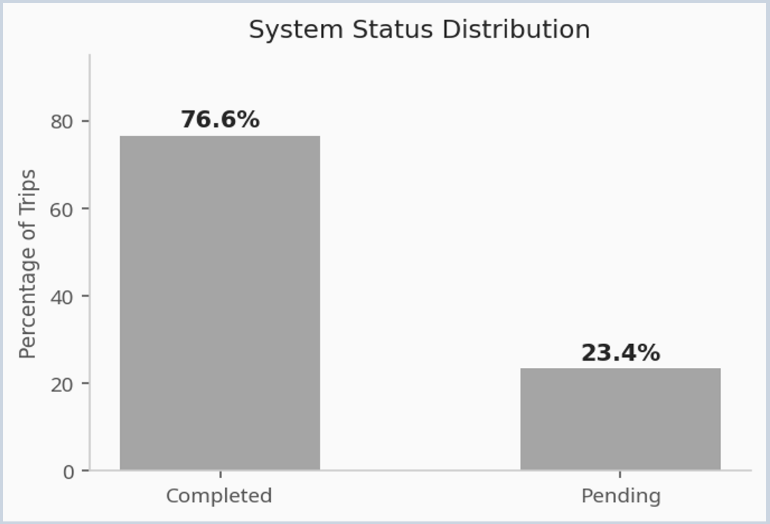
Partially reliable — manual review recommended

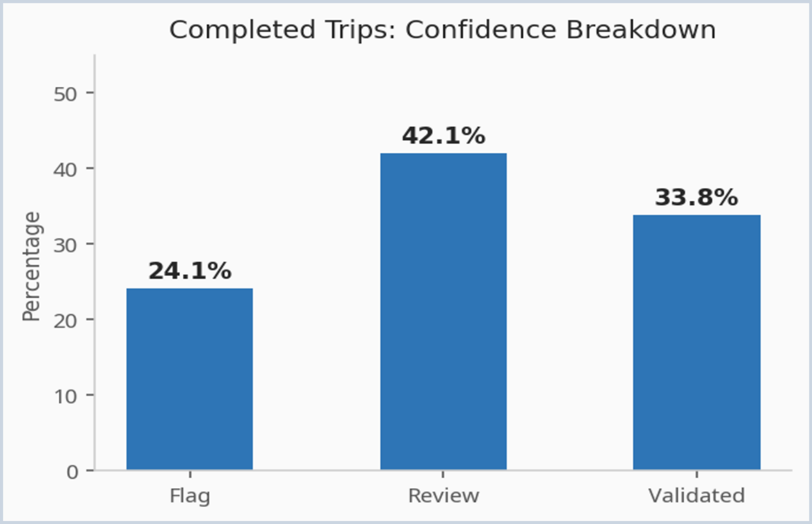
VALIDATED > 7.5

Trustworthy — anomalies reflect real driver behavior

System Status vs Confidence in Execution

We add a confidence layer to evaluate trust





Only 34% of trips are validated | ~66% require review or cannot be verified

COMPLETION ≠ RELIABLE EXECUTION